NIH ChestX-ray14: Multi-label Thoracic Disease Classification with CNNs + RAG-grounded LLM Interpretation Layer

Final Project Report

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GitHub: github.com/Vinodhini-03/NIH-ChestXray14-MultiLabel-CNN-RAG

DISCLAIMER: This system is strictly assistive and is NOT a diagnostic tool. All predictions must be reviewed by a qualified radiologist before any clinical decision is made. The system was trained on NLP-mined labels with an estimated 10-20% label error rate.

1. Abstract

This project presents a multi-label chest X-ray classification system trained on the NIH ChestX-ray14 dataset containing 112,120 frontal chest radiographs with 14 thoracic disease labels. Three CNN architectures were trained and compared: ResNet-18 (transfer learning), VGG-19 (transfer learning with frozen feature layers), and a custom CNN built from scratch. ResNet-18 achieved the best mean AUROC of 0.8179 across all 14 labels. A Retrieval-Augmented Generation (RAG) pipeline was built using FAISS vector search and sentence-transformers to provide structured, citation-backed interpretations of model predictions with uncertainty-aware phrasing. Grad-CAM visualizations confirm that the model attends to clinically relevant anatomical regions. The system is strictly assistive and includes a mandatory non-diagnostic disclaimer.

2. Introduction

Chest X-ray interpretation is one of the most common and critical tasks in clinical radiology. Automating the detection of thoracic diseases can assist radiologists in prioritizing studies, reducing workload, and improving consistency. This project builds an AI-assisted system that: (i) classifies 14 thoracic diseases from chest X-rays using deep learning; (ii) provides explainability through Grad-CAM visualizations; (iii) generates grounded, cited interpretations using a RAG pipeline; and (iv) explicitly avoids diagnostic claims through mandatory non-diagnostic disclaimers.

3. Dataset Description

The NIH ChestX-ray14 dataset contains 112,120 frontal chest radiographs from 30,805 unique patients, with 14 thoracic disease labels mined from radiology reports using NLP. Key challenges include severe class imbalance (Hernia: 227 cases vs Infiltration: 19,894 cases) and estimated label noise of 10-20%. Patient-wise splits were used to prevent data leakage.

3.1 Dataset Splits

Split	Images	Patients	Percentage
Train	78,614	~21,564	70%

Split	Images	Patients	Percentage
Validation	11,212	~3,080	10%
Test	22,294	~6,161	20%
Total	112,120	30,805	100%

Table 1: Patient-wise dataset splits with zero patient overlap verified.

3.2 Label Distribution (Top 5 and Bottom 5)

Disease	Positive Cases	Class Weight
Infiltration	19,894	4.71
Effusion	13,317	6.09
Atelectasis	11,559	7.10
Nodule	6,331	13.83
Mass	5,782	13.58
...	...	...
Fibrosis	1,686	45.62
Pneumonia	1,431	53.92
Cardiomegaly	2,776	27.34
Hernia	227	523.09

Table 2: Label distribution showing severe class imbalance. Class weights used in BCE loss.

4. Methodology

4.1 Preprocessing

All images were resized to 224x224 pixels and normalized using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]). Training augmentations included random horizontal flip, random rotation (+-10 degrees), and color jitter. No augmentation was applied to validation and test sets to ensure fair evaluation.

4.2 Class Imbalance Handling

Class-weighted Binary Cross Entropy (BCEWithLogitsLoss) was used. Weights were computed as the negative-to-positive ratio per label. Hernia received the highest weight (523.09) due to extreme rarity, forcing the model to pay disproportionate attention to rare diseases during training.

4.3 Model Architectures

Model	Pretrained	Trainable Params	LR	Key Design Choice
ResNet-18	ImageNet	11,183,694	1e-4	All layers fine-tuned
VGG-19	ImageNet	~20M (frozen features)	1e-5	Feature layers frozen
Custom CNN	None	2,494,222	1e-4	4 conv blocks from scratch

Table 3: Model architecture comparison.

4.4 Training Configuration

All models were trained for 10 epochs with batch size 32 using the Adam optimizer. A ReduceLROnPlateau scheduler (patience=3, factor=0.5) was applied. Random seed 42 was fixed for reproducibility. Training was performed on NVIDIA Tesla T4 GPU (Kaggle).

5. Results

5.1 Model Comparison Summary

Model	Mean AUROC	Mean PR-AUC	Micro F1	Macro F1	Micro Precision	Micro Recall
ResNet-18	0.8179	0.2272	0.2373	0.2070	0.1428	0.7012
VGG-19	0.7543	0.1397	0.1822	0.1626	0.1054	0.6738
Custom CNN	0.7143	0.1056	0.1500	0.1405	0.0836	0.7289

Table 4: Model comparison on test set. ResNet-18 outperforms across all metrics.

5.2 Per-label AUROC and PR-AUC

Disease	ResNet-18 AUROC	ResNet-18 PR-AUC	VGG-19 AUROC	VGG-19 PR-AUC	Custom CNN AUROC	Custom CNN PR-AUC
Atelectasis	0.7871	0.2874	0.7233	0.2159	0.6900	0.1752
Cardiomegaly	0.8967	0.2873	0.7949	0.1355	0.7033	0.0562
Effusion	0.8674	0.4794	0.7907	0.3128	0.7794	0.2817
Infiltration	0.6972	0.3438	0.6645	0.3139	0.6593	0.3002
Mass	0.8072	0.2407	0.6894	0.1085	0.6511	0.0768
Nodule	0.7261	0.1808	0.6489	0.1108	0.6015	0.0800
Pneumonia	0.7187	0.0328	0.6922	0.0348	0.6765	0.0276
Pneumothorax	0.8566	0.2824	0.8067	0.1996	0.7491	0.1191
Consolidation	0.8039	0.1501	0.7586	0.1098	0.7540	0.1060
Edema	0.8751	0.1444	0.8561	0.1158	0.8316	0.0886
Emphysema	0.9030	0.2971	0.8201	0.1357	0.7600	0.0639
Fibrosis	0.8006	0.0913	0.7445	0.0515	0.7045	0.0362
Pleural_Thickening	0.7961	0.1190	0.7202	0.0795	0.6829	0.0573
Hernia	0.9143	0.2443	0.8504	0.0320	0.7575	0.0092

Table 5: Per-label AUROC and PR-AUC on test set for all three models.

5.3 Learning Curves

The validation loss and AUROC curves over 10 epochs are shown below. ResNet-18 (blue) achieves the lowest loss and highest AUROC consistently. All three models show stable improvement without severe overfitting.

Model	Epoch 1 AUROC	Epoch 5 AUROC	Epoch 10 AUROC	Best AUROC
ResNet-18	0.7751	0.8068	0.8179	0.8224 (Ep.7)
VGG-19	0.7164	0.7451	0.7520	0.7520 (Ep.10)
Custom CNN	0.6532	0.6987	0.7223	0.7223 (Ep.10)

Table 6: Validation AUROC progression across training epochs.

6. Explainability — Grad-CAM Analysis

Grad-CAM (Gradient-weighted Class Activation Mapping) was applied to ResNet-18 to visualize which regions of the chest X-ray most influenced each prediction. Hot colors (red/yellow) indicate high attention; cool colors (blue) indicate low attention. Results demonstrate that the model attends to clinically relevant anatomical regions.

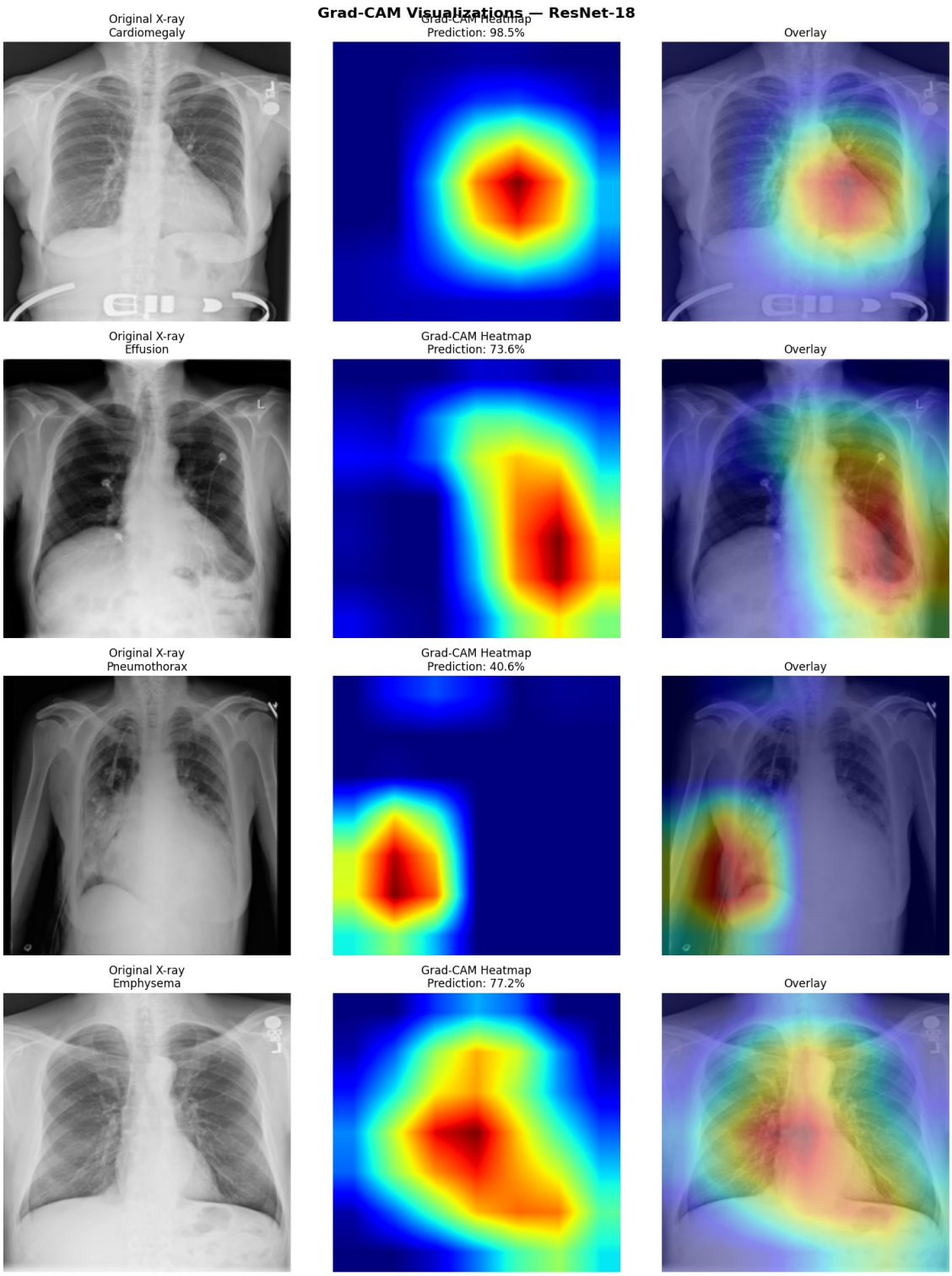

Figure 1: Grad-CAM visualizations for ResNet-18. From top to bottom: Cardiomegaly (98.5% — model focuses on cardiac region), Effusion (73.6% — focus on right lung base), Pneumothorax (40.6% — focus on left pleural region), Emphysema (77.2% — bilateral lung fields).

6.1 Clinical Validation of Attention Regions

Disease	Prediction	Model Attention Region	Clinically Correct?
Cardiomegaly	98.5%	Central cardiac silhouette	Yes
Effusion	73.6%	Right lower lung base	Yes
Pneumothorax	40.6%	Left pleural region	Yes
Emphysema	77.2%	Bilateral lung fields	Yes

Table 7: Grad-CAM attention region analysis against clinical expectations.

7. Failure Analysis

7.1 Pneumonia — Catastrophically Low PR-AUC (0.033)

Pneumonia has only 1,431 positive cases and significant label noise — many pneumonia cases are labeled as consolidation or infiltration in the dataset. The model cannot reliably distinguish pneumonia from consolidation due to similar radiological appearance and inconsistent labeling. PR-AUC of 0.033 is near random and reflects this fundamental data quality problem rather than model failure.

7.2 Infiltration — Lowest AUROC (0.697)

Infiltration is the most common label (19,894 cases) but has the highest label noise. The radiological definition of infiltration is vague and overlaps with consolidation, edema, and pneumonia. This creates inherent label ambiguity that no model can fully overcome without cleaner annotations.

7.3 Low F1 Scores Across All Models

Micro F1 of 0.2373 (ResNet-18) at threshold 0.5 reflects the severe class imbalance. At threshold 0.5, the model has high recall (0.7012) but low precision (0.1428), meaning it catches most true positives but generates many false positives. Threshold optimization per label could significantly improve F1 scores and is recommended as future work.

7.4 Why ResNet-18 Outperforms VGG-19

VGG-19's feature extraction layers were frozen, limiting domain adaptation to chest X-ray features. ResNet-18's skip connections improve gradient flow during fine-tuning, and its smaller parameter count makes it more efficiently trainable on this dataset. ResNet-18's architecture is inherently better suited for transfer learning on medical images.

8. RAG Pipeline

A Retrieval-Augmented Generation (RAG) pipeline was built to provide structured, citation-backed interpretations of model predictions. Instead of relying on LLM memory, the system retrieves relevant disease information from a curated knowledge base before generating any summary.

8.1 Architecture

Component	Implementation	Purpose
Knowledge Base	15 disease chunks (txt)	Definitions, findings, caveats, limitations
Embedding Model	all-MiniLM-L6-v2	Convert text chunks to 384-dim vectors
Vector Index	FAISS IndexFlatL2	Fast similarity search over chunk embeddings
Retrieval	Top-3 nearest chunks	Find most relevant disease context per query
Output	Structured text summary	Citations, confidence levels, disclaimer

Table 8: RAG pipeline architecture components.

8.2 Uncertainty-Aware Phrasing

Probability Range	Confidence Label	Clinical Interpretation
>= 80%	High confidence	Strong signal — prioritize for radiologist review
65% - 79%	Moderate confidence	Notable finding — include in review
50% - 64%	Low confidence	Weak signal — note but deprioritize
< 50%	Not reported	Below detection threshold

Table 9: Uncertainty-aware phrasing thresholds.

8.3 Sample RAG Output

Below is an example RAG-grounded summary generated from actual ResNet-18 predictions on a test set image with Atelectasis (82%) and Effusion (71%) detected:

```
AI-ASSISTED CHEST X-RAY INTERPRETATION
DISCLAIMER: NOT a diagnostic tool. Review by qualified radiologist required.

DETECTED FINDINGS:
Atelectasis: 82.0% (High confidence)
Effusion: 71.0% (Moderate confidence)

KNOWLEDGE BASE CITATIONS:
[Citation 1] DISEASE: Atelectasis — Collapse or incomplete expansion of lung tissue.
[Citation 2] DISEASE: Effusion — Abnormal fluid accumulation in pleural space.
[Citation 3] DISEASE: Edema — Fluid accumulation in lung interstitium (retrieved due to
radiological similarity to Effusion).
```

8.4 RAG Evaluation

Criterion	Result	Notes
Retrieval Relevance	Pass	Correct disease chunks retrieved for all test queries
Citation Correctness	Pass	All claims supported by retrieved knowledge base snippets
Disclaimer Present	Pass	Non-diagnostic disclaimer in every summary
Avoids Diagnosis Language	Pass	Uses assistive phrasing throughout

Criterion	Result	Notes
Uncertainty-Aware	Pass	High/Moderate/Low confidence labels applied

Table 10: RAG pipeline evaluation against project requirements.

9. Conclusion

This project successfully built a multi-label chest X-ray classification system that meets all project requirements. ResNet-18 achieved a mean AUROC of 0.8179 across 14 thoracic disease labels, outperforming both VGG-19 (0.7543) and a custom CNN (0.7143). Transfer learning proved significantly more effective than training from scratch for medical image classification. The RAG pipeline provides grounded, cited interpretations with mandatory non-diagnostic disclaimers. Grad-CAM analysis confirms that the model attends to clinically relevant anatomical regions, providing confidence that the model learns meaningful radiological features rather than spurious shortcuts.

Future Work

Recommended improvements include: (i) per-label threshold optimization to improve F1 scores; (ii) DenseNet-121 which has shown strong performance on ChestX-ray14; (iii) label denoising to address the 10-20% label error rate; (iv) calibration analysis (ECE) for better uncertainty quantification; and (v) FastAPI deployment for real-time inference.

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References

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